

Course: B. Tech.

Branch : Mechanical Engineering **Semester :** VI

Subject Code & Name: BTMC602 Machine Design II

Max Marks: 60

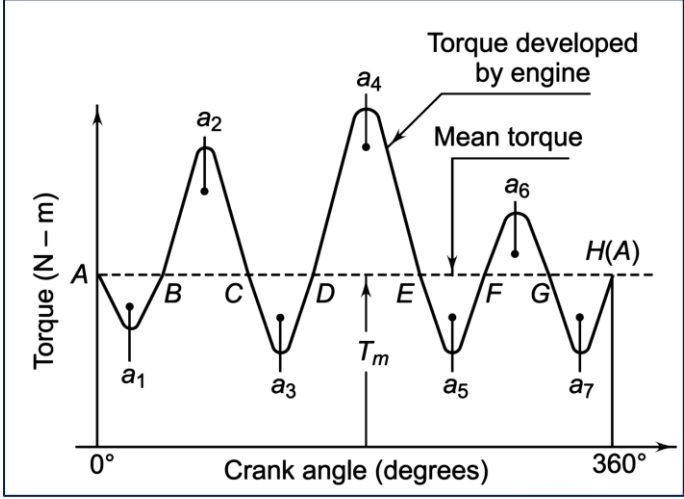
Date: 14/07/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	A single-row deep groove ball bearing is subjected to a pure radial force of 3 kN from a shaft that rotates at 600 rpm. The expected life L_{10h} of the bearing is 30,000 h. The minimum acceptable diameter of the shaft is 40 mm. Select a suitable ball bearing for this application from manufacturer's catalogue.	L3 Apply CO1 CO2	6
B)	Explain a. Dynamic Load Carrying Capacity and b. Equivalent dynamic load for a roller bearing	L2 Understand CO1 CO2	6
C)	A single-row deep groove ball bearing has a dynamic load capacity of 40500 N and operates on the following work cycle: (i) radial load of 5000 N at 500 rpm for 25% of the time; (ii) radial load of 10000 N at 700 rpm for 50% of the time; and (iii) radial load of 7000 N at 400 rpm for the remaining 25% of the time. Calculate the expected life of the bearing in hours.	L3 Apply CO1 CO2	6
Q.2	Solve Any Two of the following.		12
A)	It is required to design a pair of spur gears with 20° full-depth involute teeth based on the Lewis equation. The velocity factor is to be used to account for dynamic load. The pinion shaft is connected to a 10 kW, 1440 rpm motor. The starting torque of the motor is 150% of the rated torque. The speed reduction is 4:1. The pinion as well as the gear is made of plain carbon steel 40C8 ($S_{ut} = 600 \text{ N/mm}^2$). The factor of safety can be taken as 1.5. Considering pitch line velocity 5 m/s to begin, calculate 1. No. of teeth on pinion and gear 2. Module based on beam strength 3. PCD of pinion and gear 4. Effective load in gear tooth using corrected pitch line velocity 5. Beam strength	L3 Apply CO4 CO5	6
B)	A pair of parallel helical gears consists of a 20 teeth pinion meshing with a 100 teeth gear. The pinion rotates at 720 rpm. The normal pressure angle is 20° , while the helix angle is 25° . The face width is 40 mm and the normal module is 4 mm. The pinion as well as the gear is made of steel 40C8 ($S_{ut} = 600 \text{ N/mm}^2$) and heat treated to a surface hardness of 300 BHN. Calculate beam strength and wear strength of helical gears.	L3 Apply CO4 CO5	6

C)	Differentiate between Spur Gears and Helical Gears.	L2 Understand CO4 CO5	6
Q. 3	Solve Any Two of the following.		12
A)	A pair of bevel gears, with 20° pressure angle, consists of a 20 teeth pinion meshing with a 30 teeth gear. The module is 4 mm, while the face width is 20 mm. The material for the pinion and gear is steel 50C4 (Sut = 750 N/mm ²). The gear teeth are lapped and ground (Class-3) and the surface hardness is 400 BHN. The pinion rotates at 500 rpm and receives 2.5 kW power from the electric motor. The starting torque of the motor is 150% of the rated torque. Determine 1. Pitch angle 2. Virtual no. of teeth on pinion 3. Lewis Form Factor 4. PCD of pinion and gear 5. Cone Distance Ao 6. Bean Strength	L3 Apply CO4 CO5	6
B)	Explain advantages and disadvantages of worm gears as compared to other gears.	Level 2 Understand CO4 CO5	6
C)	Explain following terminologies of worm gears. a. Axial Pitch b. Lead c. Lead angle	Level 2 Understand CO4 CO5	6
Q.4	Solve Any Two of the following.		12
A)		Level 3 Apply	6
Demonstrate the concept & mathematical equation of maximum fluctuation of energy and coefficient of fluctuation of energy by using this diagram.			
B)	A chain drive is used in a special purpose vehicle. The vehicle is run by a 30kW rotary engine. There is a separate mechanical drive from the engine shaft to the intermediate shaft. The driving sprocket is fixed to this intermediate shaft. The efficiency of the drive between the engine and the intermediate shafts is 90%. The driving sprocket has 17 teeth and it rotates at 300 rpm. The driven sprocket rotates at 100 rpm. Assume moderate shock conditions and select a suitable four-strand chain for this drive. Use Given data from tables for selection of various factors.	Level 3 Apply CO5	6

C)	It is required to select a flat belt drive for a compressor running at 720 rpm, which is driven by a 25 kW, 1440 rpm motor. Space is available for a centre distance of 3 m. The belt is open-type. Use belt velocity as 18 m/s.	L3 Apply CO5	6
Q. 5	Solve Any Two of the following.		12
A)	A pivoted double-block brake, has two shoes, which subtend an angle (2θ) of 100° . The diameter of the brake drum is 500 mm and the width of the friction lining is 100 mm. The coefficient of friction is 0.2 and the maximum intensity of pressure between the lining and the brake drum is 0.5 N/mm^2 . The pivot of each shoe is located in such a manner that the moment of the frictional force on the shoe is zero. Calculate: (i) the distance of the pivot from the axis of the brake drum (ii) the torque capacity of each shoe (iii) the reactions at the pivot	L3 Apply CO3	6
B)	An automotive plate clutch consists of two pairs of contacting surfaces with an asbestos friction lining. The torque transmitting capacity of the clutch is 550 N-m. The coefficient of friction is 0.25 and the permissible intensity of pressure is 0.5 N/mm^2 . Due to space limitations, the outer diameter of the friction disk is fixed as 250 mm. Calculate (i) the inner diameter of the friction disk; and (ii) the spring force required to keep the clutch in an engaged position. Use uniform wear theory.	L3 Apply CO3	6
C)	A cone clutch with asbestos friction lining transmits 30 kW power at 500 rpm. The coefficient of friction is 0.2 and the permissible intensity of pressure is 0.35 N/mm^2 . The semi-cone angle α is 12.5° . The outer diameter is fixed as 300 mm from space limitations. Assuming uniform wear theory, calculate: (i) the inner diameter; (ii) the face width of the friction lining; and (iii) the force required to engage the clutch.	L3 Apply CO3	6

***** End *****

Table 17.3 <i>Values of the Lewis form factor Y for 20° full-depth involute system</i>					
<i>z</i>	<i>Y</i>	<i>z</i>	<i>Y</i>	<i>z</i>	<i>Y</i>
15	0.289	27	0.348	55	0.415
16	0.295	28	0.352	60	0.421
17	0.302	29	0.355	65	0.425
18	0.308	30	0.358	70	0.429
19	0.314	32	0.364	75	0.433
20	0.320	33	0.367	80	0.436
21	0.326	35	0.373	90	0.442
22	0.330	37	0.380	100	0.446
23	0.333	39	0.386	150	0.458
24	0.337	40	0.389	200	0.463
25	0.340	45	0.399	300	0.471
26	0.344	50	0.408	Rack	0.484

Table 15.5 Dimensions and Static and Dynamic Load Capacities of Single Row Deep Groove Ball Bearing

<i>Principal dimensions (mm)</i>			<i>Basic load ratings (N)</i>		<i>Designation</i>
<i>d</i>	<i>D</i>	<i>B</i>	<i>C</i>	<i>C₀</i>	
40	52	7	4160	3350	61808
	68	9	13300	7800	16008
	68	15	16800	9300	6008
	80	18	30700	16600	6208
	90	23	41000	22400	6308
	110	27	63700	36500	6408
45	58	7	6050	3800	61809
	75	10	15600	9300	16009
	75	16	21200	12200	6009
	85	19	33200	18600	6209
	100	25	52700	30000	6309
	120	29	76100	45500	6409
50	65	7	6240	4250	61810
	80	10	16300	10000	16010
	80	16	21600	13200	6010
	90	20	35100	19600	6210
	110	27	61800	36000	6310
	130	31	87100	52000	6410

Design Data for Roller Chains

Table 14.2 Power rating of simple roller chain

Pinion speed (rpm)	Power (kW)								
	06 B	08A	08 B	10A	10 B	12A	12 B	16A	16 B
50	0.14	0.28	0.34	0.53	0.64	0.94	1.07	2.06	2.59
100	0.25	0.53	0.64	0.98	1.18	1.74	2.01	4.03	4.83
200	0.47	0.98	1.18	1.83	2.19	3.40	3.75	7.34	8.94
300	0.61	1.34	1.70	2.68	3.15	4.56	5.43	11.63	13.06
500	1.09	2.24	2.72	4.34	5.01	7.69	8.53	16.99	20.57
700	1.48	2.95	3.66	5.91	6.71	10.73	11.63	23.26	27.73
1000	2.03	3.94	5.09	8.05	8.97	14.32	15.65	28.63	34.89
1400	2.73	5.28	6.81	11.18	11.67	14.32	18.15	18.49	38.47
1800	3.44	6.98	8.10	8.05	13.03	10.44	19.85	—	—
2000	3.80	6.26	8.67	7.16	13.49	8.50	20.57	—	—

Table 14.3 Service factor (K_s)

Type of driven load	Type of input power		
	IC engine with hydraulic drive	Electric motor	IC engine with mechanical drive
(i) Smooth: agitator, fan, light conveyor	1.0	1.0	1.2
(ii) Moderate shock: machine tools, crane, heavy conveyor, food mixer, grinder	1.2	1.3	1.4
(iii) Heavy shock: punch press, hammer mill, reciprocating conveyor, rolling mill drive	1.4	1.4	1.7

Table 14.4 Multiple strand factor (K_1)

Number of strands	K_1
1	1.0
2	1.7
3	2.5
4	3.3
5	3.9
6	4.6

Table 14.5 Tooth correction factor (K_2)

Number of teeth on the driving sprocket	K_2
15	0.85
16	0.92
17	1.00
18	1.05
19	1.11
20	1.18
21	1.26
22	1.29
23	1.35
24	1.41
25	1.46
30	1.73

Data Tables for Belt Design

Table 13.1 Load correction factor (F_a)

Type of load	F_a
(i) Normal load	1.0
(ii) Steady load, e.g., centrifugal pumps, fans, light machine tools, conveyors	1.2
(iii) Intermittent load, e.g., heavy duty fans, blowers, compressors, reciprocating pumps, line shafts, heavy-duty machines	1.3
(iv) Shock load, e.g., vacuum pumps, rolling mills, hammers, grinders	1.5

Table 13.2 Arc of contact factor (F_d)

α_s (Deg.)	120	130	140	150	160	170	180	190	200
F_d	1.33	1.26	1.19	1.13	1.08	1.04	1.00	0.97	0.94

HI-SPEED	0.0118 kW per mm width per ply
FORT	0.0147 kW per mm width per ply

The standard widths of belts (in mm)

3-Ply	25	40	50	63	76					
4-Ply	40	44	50	63	76	90	100	112	125	152
5-Ply	76	100	112	125	152					
6-Ply	112	125	152	180	200					

Table 13.3 Minimum pulley diameters for given belt speeds and belt plies (mm)

No. of plies	Maximum belt speed (m/s)				
	10	15	20	25	30
3	90	100	112	140	180
4	140	160	180	200	250
5	200	224	250	315	355
6	250	315	355	400	450
7	355	400	450	500	560
8	450	500	560	630	710
9	560	630	710	800	900
10	630	710	800	900	1000